Forces & Equilibrium

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Resolving Forces in Two Dimensions

Solve the inequality $x^2 + 17x + 72 > 0$.

The quadratic factorises as $(x + 9)(x + 8)$ with roots -9 and -8; since the parabola opens upward, the expression is > 0 outside the roots, giving $x < -9$ or $x > -8$.

2 [Foundational] • Target: Friction & Limiting Equilibrium

Solve the inequality $x^2 + x - 20 > 0$.

The quadratic factorises as $(x - 4)(x + 5)$ with roots -5 and 4; since the parabola opens upward, the expression is > 0 outside the roots, giving $x < -5$ or $x > 4$.

3 [Foundational] • Target: Inclined Planes & Equilibrium

Solve the inequality $x^2 + 7x - 8 > 0$.

The quadratic factorises as $(x + 8)(x - 1)$ with roots -8 and 1; since the parabola opens upward, the expression is > 0 outside the roots, giving $x < -8$ or $x > 1$.

4 [Foundational] • Target: Lami's Theorem & Connected Systems

Solve the inequality $x^2 - 4x - 5 > 0$.

The quadratic factorises as $(x + 1)(x - 5)$ with roots -1 and 5; since the parabola opens upward, the expression is > 0 outside the roots, giving $x < -1$ or $x > 5$.

5 [Foundational] • Target: Resolving Forces in Two Dimensions

Solve the inequality $x^2 - 8x + 12 > 0$.

The quadratic factorises as $(x - 6)(x - 2)$ with roots 2 and 6; since the parabola opens upward, the expression is > 0 outside the roots, giving $x < 2$ or $x > 6$.

6 [Foundational] • Target: Friction & Limiting Equilibrium

Solve the inequality $x^2 + 6x > 0$.

The quadratic factorises as $(x + 6)x$ with roots -6 and 0; since the parabola opens upward, the expression is > 0 outside the roots, giving $x < -6$ or $x > 0$.

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7 [Foundational] • Target: Inclined Planes & Equilibrium

A particle of mass m kg is placed on a rough inclined plane. The plane is inclined at an angle α to the horizontal, where $\tan\alpha = \frac{3}{4}$. The particle is on the point of sliding down the plane. The coefficient of static friction between the particle and the plane is μ_s. Given that the acceleration due to gravity is $g = 9.8$ m/s^2, find the value of μ_s.

Final Answer: $\frac{3}{4}$. When a particle is on the point of sliding down an inclined plane, the angle of inclination α is equal to the angle of friction ϕ, where $\tan\phi = \mu_s$. Given that $\tan\alpha = \frac{3}{4}$, and the particle is on the point of sliding, we have $\alpha = \phi$. Therefore, $\mu_s = \tan\phi = \tan\alpha = \frac{3}{4}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Lami's Theorem & Connected Systems

A block rests on a rough horizontal surface with normal reaction $N = 42$ N. The coefficient of friction is $\mu = 7/20$. Find the limiting frictional force.

Final Answer: 147/10. Applying Forces & Equilibrium principles, $F = \mu N = 7/20 \times 42 = 147/10$ N Therefore, the verified correct option is 147/10.

9 [Intermediate] • Target: Resolving Forces in Two Dimensions

A block rests on a rough horizontal surface with normal reaction $N = 55$ N. The coefficient of friction is $\mu = 13/20$. Find the limiting frictional force.

Final Answer: 143/4. Applying Forces & Equilibrium principles, $F = \mu N = 13/20 \times 55 = 143/4$ N Therefore, the verified correct option is 143/4.

10 [Intermediate] • Target: Friction & Limiting Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 16$ N. The coefficient of friction is $\mu = 11/20$. Find the limiting frictional force.

Final Answer: 44/5. Applying Forces & Equilibrium principles, $F = \mu N = 11/20 \times 16 = 44/5$ N Therefore, the verified correct option is 44/5.

11 [Intermediate] • Target: Inclined Planes & Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 15$ N. The coefficient of friction is $\mu = 7/10$. Find the limiting frictional force.

Final Answer: 21/2. Applying Forces & Equilibrium principles, $F = \mu N = 7/10 \times 15 = 21/2$ N Therefore, the verified correct option is 21/2.

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12 [Intermediate] • Target: Lami's Theorem & Connected Systems

A block rests on a rough horizontal surface with normal reaction $N = 45$ N. The coefficient of friction is $\mu = 7/10$. Find the limiting frictional force.

Final Answer: 63/2. Applying Forces & Equilibrium principles, $F = \mu N = 7/10 \times 45 = 63/2$ N Therefore, the verified correct option is 63/2.

13 [Intermediate] • Target: Resolving Forces in Two Dimensions

A block rests on a rough horizontal surface with normal reaction $N = 12$ N. The coefficient of friction is $\mu = 3/10$. Find the limiting frictional force.

Final Answer: 18/5. Applying Forces & Equilibrium principles, $F = \mu N = 3/10 \times 12 = 18/5$ N Therefore, the verified correct option is 18/5.

14 [Intermediate] • Target: Friction & Limiting Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 18$ N. The coefficient of friction is $\mu = 7/10$. Find the limiting frictional force.

Final Answer: 63/5. Applying Forces & Equilibrium principles, $F = \mu N = 7/10 \times 18 = 63/5$ N Therefore, the verified correct option is 63/5.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Inclined Planes & Equilibrium • Cross-Lesson: 9.01 Circular Measure

A block rests on a rough horizontal surface with normal reaction $N = 16$ N. The coefficient of friction is $\mu = 7/20$. Find the limiting frictional force.

Final Answer: 28/5. Applying Forces & Equilibrium principles, $F = \mu N = 7/20 \times 16 = 28/5$ N Therefore, the verified correct option is 28/5.

16 [Challenging] • Target: Lami's Theorem & Connected Systems • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A block rests on a rough horizontal surface with normal reaction $N = 55$ N. The coefficient of friction is $\mu = 9/20$. Find the limiting frictional force.

Final Answer: 99/4. Applying Forces & Equilibrium principles, $F = \mu N = 9/20 \times 55 = 99/4$ N Therefore, the verified correct option is 99/4.

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17 [Challenging] • Target: Resolving Forces in Two Dimensions • Cross-Lesson: 10.01 Series & Binomial Theorem

A block rests on a rough horizontal surface with normal reaction $N = 55$ N. The coefficient of friction is $\mu = 3/20$. Find the limiting frictional force.

Final Answer: 33/4. Applying Forces & Equilibrium principles, $F = \mu N = 3/20 \times 55 = 33/4$ N Therefore, the verified correct option is 33/4.

18 [Challenging] • Target: Friction & Limiting Equilibrium • Cross-Lesson: 11.01 Kinematics

A block rests on a rough horizontal surface with normal reaction $N = 55$ N. The coefficient of friction is $\mu = 7/20$. Find the limiting frictional force.

Final Answer: 77/4. Applying Forces & Equilibrium principles, $F = \mu N = 7/20 \times 55 = 77/4$ N Therefore, the verified correct option is 77/4.

19 [Challenging] • Target: Inclined Planes & Equilibrium • Cross-Lesson: 13.01 Newton's Laws

A block rests on a rough horizontal surface with normal reaction $N = 30$ N. The coefficient of friction is $\mu = 1/4$. Find the limiting frictional force.

Final Answer: 15/2. Applying Forces & Equilibrium principles, $F = \mu N = 1/4 \times 30 = 15/2$ N Therefore, the verified correct option is 15/2.

20 [Challenging] • Target: Lami's Theorem & Connected Systems • Cross-Lesson: 14.01 Energy, Work & Power

A block rests on a rough horizontal surface with normal reaction $N = 50$ N. The coefficient of friction is $\mu = 1/4$. Find the limiting frictional force.

Final Answer: 25/2. Applying Forces & Equilibrium principles, $F = \mu N = 1/4 \times 50 = 25/2$ N Therefore, the verified correct option is 25/2.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Resolving Forces in Two Dimensions

A force of magnitude 50 N acts at an angle of 30° above the horizontal. Find its horizontal and vertical components.

a. **Horizontal:** 43.3 N, **Vertical:** 25.0 N ✓

b. Horizontal: 25.0 N, Vertical: 43.3 N

c. Horizontal: 35.4 N, Vertical: 35.4 N

d. Horizontal: 50.0 N, Vertical: 0.0 N

$F_x = 50\cos(30^\circ) = 50(\sqrt{3}/2) \approx 43.3$ N. $F_y = 50\sin(30^\circ) = 50(1/2) = 25.0$ N.

22 [Foundational] • Target: Resolving Forces in Two Dimensions

Two forces $F_1 = [8,\ -3]$ N and $F_2 = [-5, 7]$ N act on a particle. What is the magnitude of the resultant force?

a. 5 N ✓

b. 7 N

c. $\sqrt{41}$ N

d. 12 N

Resultant $\mathbf{R} = F_1 + F_2 = [8 - 5,\ -3 + 7] = [3, 4]$ N. Magnitude $|\mathbf{R}| = \sqrt{3^2 + 4^2} = 5$ N.

23 [Foundational] • Target: Resolving Forces in Two Dimensions

A particle is held in equilibrium by three coplanar forces: P acting East, Q acting North, and a force of 13 N acting at an angle θ South of West where $\tan\theta = 5/12$. Find the magnitude of P.

a. 12 N ✓

b. 5 N

c. 13 N

d. 8 N

For equilibrium, horizontal components balance: $P = 13\cos\theta$. Since $\tan\theta = 5/12$, $\cos\theta = 12/13$, so $P = 13(12/13) = 12$ N.

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24 [Foundational] • Target: Friction & Limiting Equilibrium

A block of mass 6 kg rests on a rough horizontal table where the coefficient of friction is $\mu = 0.35$. Taking $g = 10$ m/s^2, what is the maximum frictional force that can act on the block?

a. 21 N ✓

b. 60 N

c. 18 N

d. 24 N

Normal reaction $R = mg = 6(10) = 60$ N. Maximum static friction $F_{\max} = \mu R = 0.35(60) = 21$ N.

25 [Foundational] • Target: Friction & Limiting Equilibrium

A horizontal force of 15 N is applied to the 6 kg block in the previous question (where $F_{\max} = 21$ N). What is the actual frictional force acting on the block?

a. 15 N ✓

b. 21 N

c. 0 N

d. 6 N

Since the applied force (15 N) is less than the limiting friction (21 N), the block remains at rest and friction balances the applied force: $F = 15$ N.

26 [Foundational] • Target: Friction & Limiting Equilibrium

A particle of mass 4 kg is placed on a rough plane inclined at 30° to the horizontal. If the particle is on the point of slipping down the plane, find the coefficient of friction μ.

a. $\mu = \tan(30^\circ) = \frac{\sqrt{3}}{3} \approx 0.577$ ✓

b. $\mu = \sin(30^\circ) = 0.5$

c. $\mu = \cos(30^\circ) \approx 0.866$

d. $\mu = 0.25$

At limiting equilibrium on an incline, $\mu = \tan\alpha$. For $\alpha = 30^\circ$, $\mu = \tan(30^\circ) = 1/\sqrt{3} = \sqrt{3}/3 \approx 0.577$.

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27 [Foundational] • Target: Inclined Planes & Equilibrium

A crate of mass 10 kg is held in equilibrium on a smooth slope inclined at 30° to the horizontal by a force P acting parallel to the slope. Taking $g = 10$ m/s^2, find P.

a. 50 N ✓

b. 86.6 N

c. 100 N

d. 25 N

Resolving parallel to the slope: $P = mg\sin(30^\circ) = 10(10)(0.5) = 50$ N.

28 [Foundational] • Target: Inclined Planes & Equilibrium

For the crate in the previous question, what is the normal contact force exerted by the slope on the crate?

a. 86.6 N **(**$50\sqrt{3}$ N**)** ✓

b. 50 N

c. 100 N

d. 75 N

Resolving perpendicular to the slope: $R = mg\cos(30^\circ) = 100(\sqrt{3}/2) = 50\sqrt{3} \approx 86.6$ N.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Inclined Planes & Equilibrium

A body of mass 8 kg rests in limiting equilibrium on an inclined plane. The normal reaction is 64 N. Taking $g = 10$ m/s^2, find the angle of inclination α of the plane.

a. $\alpha = \arccos(0.8) \approx 36.9^\circ$ ✓

b. $\alpha = 30^\circ$

c. $\alpha = 45^\circ$

d. $\alpha = 60^\circ$

$R = mg\cos\alpha \Rightarrow 64 = 8(10)\cos\alpha \Rightarrow 80\cos\alpha = 64 \Rightarrow \cos\alpha = 0.8 \Rightarrow \alpha \approx 36.9^\circ$.

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30 [Intermediate] • Target: Lami's Theorem & Connected Systems

A lamp of weight $W = 60$ N is suspended by two light strings making angles of 30° and 60° with the horizontal ceiling. Find the tension in the steeper string.

a. $30\sqrt{3}$ N ≈ 52.0 N ✓

b. 30 N

c. 60 N

d. 45 N

The angles with the vertical are 60° and 30°. The two strings are perpendicular ($30^\circ + 60^\circ = 90^\circ$). Resolving along the steeper string: $T_2 = W\cos(30^\circ) = 60(\sqrt{3}/2) = 30\sqrt{3}$ N ≈ 52.0 N.

31 [Intermediate] • Target: Lami's Theorem & Connected Systems

A light inextensible string passes over a smooth peg. A mass of 5 kg hangs on one side and a mass of 3 kg hangs on the other. What downward external force must be applied to the 3 kg mass to maintain equilibrium? ($g = 10$ m/s^2)

a. 20 N ✓

b. 50 N

c. 30 N

d. 10 N

Tension in string is $T = 5g = 50$ N. On the 3 kg side: $T = 3g + F \Rightarrow 50 = 30 + F \Rightarrow F = 20$ N.

32 [Intermediate] • Target: Lami's Theorem & Connected Systems

A ring of mass m is threaded on a rough horizontal wire with coefficient of friction $\mu = 0.4$. A light string attached to the ring is pulled with tension T at an angle of 45° above the wire. Find the value of T (in terms of mg) when the ring is in limiting equilibrium.

a. $T = \dfrac{0.4mg}{\cos(45^\circ) + 0.4\sin(45^\circ)} = \dfrac{0.4mg}{1.4\frac{\sqrt{2}}{2}} = \dfrac{4\sqrt{2}}{7}mg$ ✓

b. $\dfrac{\sqrt{2}}{2}mg$

c. $0.4mg$

d. mg

Resolving vertically: $R + T\sin(45^\circ) = mg \Rightarrow R = mg - T\sin(45^\circ)$. Horizontal friction: $F = \mu R = 0.4(mg - T\sin(45^\circ))$. For limiting equilibrium: $T\cos(45^\circ) = 0.4(mg - T\sin(45^\circ)) \Rightarrow T(\cos 45^\circ + 0.4\sin 45^\circ) = 0.4mg \Rightarrow T = \frac{0.4mg}{1.4(\sqrt{2}/2)} = \frac{4\sqrt{2}}{7}mg$.

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33 [Intermediate] • Target: Resolving Forces in Two Dimensions

A block rests on a rough horizontal surface with normal reaction $N = 18$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. 46/5

b. 26/5

c. 72/5

d. **36/5 ✓**

Applying Forces & Equilibrium principles, $F = \mu N = 2/5 \times 18 = 36/5$ N Therefore, the verified correct option is 36/5.

34 [Intermediate] • Target: Friction & Limiting Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 18$ N. The coefficient of friction is $\mu = 3/5$. Find the limiting frictional force.

a. 44/5

b. 64/5

c. **54/5 ✓**

d. 108/5

Applying Forces & Equilibrium principles, $F = \mu N = 3/5 \times 18 = 54/5$ N Therefore, the verified correct option is 54/5.

35 [Intermediate] • Target: Inclined Planes & Equilibrium

A block rests on a rough horizontal surface with normal reaction $N = 15$ N. The coefficient of friction is $\mu = 1/4$. Find the limiting frictional force.

a. 7/4

b. 23/4

c. 15/2

d. **15/4 ✓**

Applying Forces & Equilibrium principles, $F = \mu N = 1/4 \times 15 = 15/4$ N Therefore, the verified correct option is 15/4.

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36 [Intermediate] • Target: Lami's Theorem & Connected Systems

A block rests on a rough horizontal surface with normal reaction $N = 50$ N. The coefficient of friction is $\mu = 3/10$. Find the limiting frictional force.

a. **15** ✓ **b.** 17

c. 30 **d.** 13

Applying Forces & Equilibrium principles, $F = \mu N = 3/10 \times 50 = 15$ N Therefore, the verified correct option is 15.

37 [Intermediate] • Target: Resolving Forces in Two Dimensions

A block rests on a rough horizontal surface with normal reaction $N = 40$ N. The coefficient of friction is $\mu = 3/10$. Find the limiting frictional force.

a. 24 **b.** 14

c. 10 **d.** **12** ✓

Applying Forces & Equilibrium principles, $F = \mu N = 3/10 \times 40 = 12$ N Therefore, the verified correct option is 12.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Friction & Limiting Equilibrium • Cross-Lesson: 11.01 Kinematics

A block rests on a rough horizontal surface with normal reaction $N = 40$ N. The coefficient of friction is $\mu = 1/2$. Find the limiting frictional force.

a. **20** ✓ **b.** 40

c. 18 **d.** 22

Applying Forces & Equilibrium principles, $F = \mu N = 1/2 \times 40 = 20$ N Therefore, the verified correct option is 20.

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39 [Challenging] • Target: Inclined Planes & Equilibrium • Cross-Lesson: 13.01 Newton's Laws

A block rests on a rough horizontal surface with normal reaction $N = 30$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. **12** ✓

b. 14

c. 24

d. 10

Applying Forces & Equilibrium principles, $F = \mu N = 2/5 \times 30 = 12$ N Therefore, the verified correct option is 12.

40 [Challenging] • Target: Lami's Theorem & Connected Systems • Cross-Lesson: 14.01 Energy, Work & Power

A block rests on a rough horizontal surface with normal reaction $N = 40$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. 14

b. 18

c. 32

d. **16** ✓

Applying Forces & Equilibrium principles, $F = \mu N = 2/5 \times 40 = 16$ N Therefore, the verified correct option is 16.

41 [Challenging] • Target: Resolving Forces in Two Dimensions • Cross-Lesson: 15.01 Representation of Data

A block rests on a rough horizontal surface with normal reaction $N = 45$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. 16

b. 20

c. **18** ✓

d. 36

Applying Forces & Equilibrium principles, $F = \mu N = 2/5 \times 45 = 18$ N Therefore, the verified correct option is 18.

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42 [Challenging] • Target: Friction & Limiting Equilibrium • Cross-Lesson: 16.01 Permutations & Combinations

A block rests on a rough horizontal surface with normal reaction $N = 30$ N. The coefficient of friction is $\mu = 3/10$. Find the limiting frictional force.

a. 18

b. 7

c. **9** ✓

d. 11

Applying Forces & Equilibrium principles, $F = \mu N = 3/10 \times 30 = 9$ N Therefore, the verified correct option is 9.

43 [Challenging] • Target: Inclined Planes & Equilibrium • Cross-Lesson: 17.01 Discrete Random Variables

A block rests on a rough horizontal surface with normal reaction $N = 10$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. **4** ✓

b. 8

c. 6

d. 2

Applying Forces & Equilibrium principles, $F = \mu N = 2/5 \times 10 = 4$ N Therefore, the verified correct option is 4.

44 [Challenging] • Target: Lami's Theorem & Connected Systems • Cross-Lesson: 18.01 Binomial Distribution

A block rests on a rough horizontal surface with normal reaction $N = 20$ N. The coefficient of friction is $\mu = 2/5$. Find the limiting frictional force.

a. 6

b. 16

c. 10

d. **8** ✓

Applying Forces & Equilibrium principles, $F = \mu N = 2/5 \times 20 = 8$ N Therefore, the verified correct option is 8.

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45 [Challenging] • Target: Resolving Forces in Two Dimensions • Cross-Lesson: 19.01 Normal Distribution

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

b. **20** ✓

c. 25

d. 22

By symmetry (each pair at 120°), the third force balances the resultant of the two equal forces: $F_3 = F_1 = 20$ N (standard 120°-120°-120° symmetric equilibrium).

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